

Hackathon Problem Statements

Regd No: 25091A32A4, 25091A32D2

Problem Statement No: 29

Elderly Fall-Risk Prediction System Create a fall-risk prediction tool using time-series classification on wearable sensor data (simulated or public datasets) capturing gait, balance, and movement patterns of elderly individuals. Flag high-risk individuals for preventive intervention like physiotherapy or home modifications. Use OOP to model sensor readings, individuals, and risk assessments as connected entities. This supports elderly care facilities and family caregivers in preventing serious fall-related injuries.

Regd No: 25091A32G1, 25091A32E2

Problem Statement No: 59

Timetable Generation System Create an automated timetable generation system using graph coloring algorithms to solve the constraint satisfaction problem of scheduling classes, teachers, and rooms without conflicts. Allow configurable constraints (teacher availability, room capacity, subject frequency) modeled through OOP classes. This solves a genuinely painful administrative task colleges face every semester, saving significant manual effort.

Regd No: 25091A3287, 25091A32J3

Problem Statement No: 73

Renewable Energy Output Predictor Build a solar/wind energy output prediction model using regression analysis on historical weather data (irradiance, wind speed, temperature) to forecast expected power generation for a given installation. Use OOP to model different renewable energy source types with distinct prediction logic. This helps energy planners and grid operators better balance renewable supply with demand, improving grid stability.

Regd No: 25091A32N3, 25091A3295

Problem Statement No: 84

Air Pollution Source Attribution System Design a pollution source attribution tool using correlation and regression analysis to statistically link AQI spikes with likely contributing sources (vehicular traffic, industrial activity, stubble burning) based on time-of-day and seasonal patterns in public datasets. Use OOP to structure pollution sources and attribution models. This provides actionable evidence for targeted pollution control policy interventions.

Regd No: 25091A32F7, 25091A32P8

Problem Statement No: 92

Personal Budget Tracker & Expense Classifier Design a personal finance tracker that uses OOP-based transaction modeling combined with basic NLP techniques to automatically categorize expenses from transaction descriptions (groceries, utilities, entertainment). Generate spending trend visualizations and savings recommendations. This promotes financial literacy and better money management among students and first-time earners.

Regd No: 25091A3211, 25091A3246

Problem Statement No: 133

Census Data Anomaly & Inconsistency Detector Build a data validation tool using statistical validation algorithms to detect inconsistencies and anomalies in census or survey datasets (impossible age values, duplicate entries, logical contradictions) before they propagate into policy-making datasets. Use OOP to model records, validation rules, and error reports. This improves the reliability of foundational government datasets used for planning.